

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **update discovery, installation, and lifecycle management features** in **Java Spring Boot** and **Python** for enterprise Windows update stack, reducing client-side update failures by **18%** through resilient error handling and comprehensive unit/integration testing across **PostgreSQL** and **Oracle** databases.
- Built **ML/AI model delivery pipeline** using **Python** data engineering and **REST API** integrations to enable seamless distribution of innovation components to validation rings and retail customers, accelerating model deployment cycles by **35%** while maintaining strict **security vulnerability assessment** and **architecture resilience standards**.
- Architected **scalable update platform monitoring and diagnostics solution** leveraging **CloudWatch**, **Splunk**, and **Dynatrace** for real-time visibility into monthly retail Windows releases; designed automated alerting and incident mitigation workflows that reduced mean-time-to-resolution (MTTR) for production issues by **42%** across multiple validation rings.
- Developed quarterly feature planning and rapid release processes in **Jira** / Agile with **Git** branching strategy and **Jenkins CI/CD pipelines**, enabling fast yet reliable shipping of new update stack code; coordinated **change-control management**, UAT validation, and cross-functional collaboration with **WSD and Windows partner teams** globally to deliver **12+ feature releases** annually with zero critical rollbacks.
- Mentored **3 junior developers and interns** through hands-on code reviews, design sessions, and architectural guidance on **C#, Java, and Python** best practices; established internal training materials and onboarding documentation that reduced new-team-member ramp-time by **40%** and improved code quality metrics.
- Implemented **security-hardened update client architecture** addressing evolving threat landscape through **AWS EC2/S3/Lambda** secure infrastructure, **Docker containerization**, and compliance frameworks; conducted **security vulnerability assessments**, remediation validation, and post-implementation triage for **50+ security patches** shipped to billions of customers with **zero critical CVEs post-release**.

SKILLS

Languages & Frameworks – C#, Java, Python, C, C++, Spring Boot, REST APIs, Flask, FastAPI.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins CI/CD, change-control management.

Databases & Data Engineering – PostgreSQL, MySQL, Oracle, SQL, Python data pipelines, ETL, data modelling, data warehousing.

Monitoring, Observability & BI – Splunk, Dynatrace, CloudWatch, Power BI, Tableau, DAX, Power Query, operational dashboards.

Agile & Collaboration – Jira, Confluence, Agile/Scrum, code reviews, cross-functional team leadership, mentoring.

Machine Learning & Analytics – scikit-learn, pandas, NumPy, R/RStudio, statistical analysis, supervised/unsupervised learning, model evaluation metrics.

EDUCATION AND TRAINING

PROJECTS

Windows Update Stack Reliability & Agentic AI Integration Platform (HCL Technologies)

- Architected end-to-end **update discovery, installation, and lifecycle orchestration system** in **Java Spring Boot** with **Python AI model inference pipelines** enabling seamless delivery of **ML/AI models and security updates** to Windows validation rings; integrated **REST APIs** across distributed microservices to support **Agentic AI and AI Orchestration** workflows, reducing update latency by **28%** and supporting **2.5B+ monthly update transactions**.
- Implemented **scalable monitoring and diagnostics fabric** using **CloudWatch, Splunk**, and custom **Power BI dashboards** to provide real-time visibility into client-side update state, failure modes, and retry patterns across **12+ validation rings**; automated incident detection and mitigation workflows that reduced **P1 retail release issues by 31%** and enabled **sub-2-hour MTTR** for critical diagnostics.
- Built **secure, resilient architecture** addressing modern security landscape threats through **AWS EC2/Lambda containerized deployment** with **Docker**, automated **security vulnerability scanning** in **Jenkins CI/CD pipelines**, and comprehensive **change-control validation**; shipped **45+ security patches** to retail with zero post-release CVEs and **99.97% availability SLA** across all update components.
- Designed quarterly **rapid feature planning and release cycle** leveraging **Jira** Agile workflows, **Git** branching strategies, and automated **Jenkins deployment automation**; coordinated cross-team collaboration with **WSD, Windows, and AI Orchestration teams** to deliver **8 major feature releases** annually while maintaining strict SLA and compliance standards for **billions of customers globally**.

SWIFT Payments Operations Dashboard & ISO 20022 Message Flow Reliability System (HCL Technologies)

- Developed production-grade **SWIFT messaging and payments operations monitoring dashboard** using **Splunk** and **CloudWatch** for a Tier-1 banking client, providing real-time visibility into **MT/MX message flows, SEPA/cross-border payment processing**, and **ISO 20022 message structure validation** across **8,000+ daily payment transactions**; achieved **99.9% uptime SLA** and reduced transaction-flow incident response time by **52%**.
- Implemented **regulatory compliance and audit trail automation** across payments operations platform, embedding **GDPR data protection** controls, transaction-level reconciliation workflows, and **PSD2/Open Banking** requirement validation into core update and monitoring logic; coordinated **change-control management** with banking domain BAs and regulators to ship **12+ compliance features** with zero regulatory deviations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)